

33. You purchase 10 bananas that weigh 50 grams each. If bananas cost \$5.50 per kilogram, how much did you pay?
34. A family-size package of ground meat costs \$15.75. The package weighs 4.5 kg. What is the cost per gram of the meat?
35. A box containing 6 identical books weighs 7.2 kg. The box weighs 600 g. What is the weight of each book in grams?
36. A 2.316 kg bag of candy is equally divided into 12 party bags. What is the weight of the candy, in grams, in each party bag?
37. A store has 450 kg of flour at the beginning of the day. At the end of the day the store has 341 kg of flour. If flour costs \$0.35 per kilogram, how much flour, in dollars, did the store sell that day?
38. A local restaurant offers lobster for \$110 per kilogram. What is the price for a lobster that weighs 450 grams?
39. A student's backpack weighs 575 grams. Their books weigh 3.5 kg. If the student's weight while wearing their backpack is 58.25 kg, how much does the student weigh in kilograms?
40. The weight of a lamb is 41 kg 340 g. What is the total weight, in kilograms, of four lambs of the same weight?

9.5 Measuring Temperature

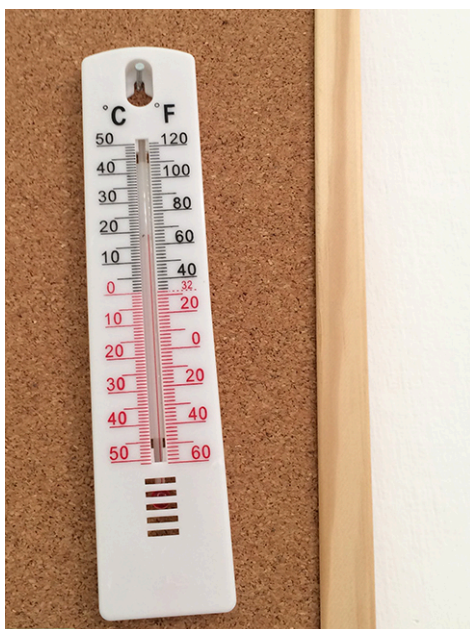


Figure 9.14 A thermometer that measures temperature in both customary and metric units. (credit: “Thermometer” by Jeff Djevdet/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Convert between Fahrenheit and Celsius.
2. Identify reasonable values for temperature applications.
3. Solve application problems involving temperature.

When you touch something and it feels warm or cold, what is that really telling you about that substance? **Temperature** is a measure of how fast atoms and molecules are moving in a substance, whether that be the air, a stove top, or an ice cube. The faster those atoms and molecules move, the higher the temperature.

In the metric system, temperature is measured using the Celsius (°C) scale. Because temperature is a condition of the physical properties of a substance, the Celsius scale was created with 100 degrees separating the point at which water freezes, 0 °C, and the point at which water boils, 100 °C. Scientifically, these are the points at which water molecules change from one state of matter to another—from solid (ice) to liquid (water) to gas (water vapor).

 When reading temperatures, it's important to look beyond the degree symbol to determine which temperature scale the units express. For example, 13 °C reads “13 degrees Celsius,” indicating that the temperature is expressed using the Celsius scale, while 13 °F reads “13 degrees Fahrenheit,” indicating that the temperature is expressed using the

Fahrenheit scale.

▶ VIDEO

[Misconceptions About Temperature \(https://openstax.org/r/Misconceptions_About_Temperature\)](https://openstax.org/r/Misconceptions_About_Temperature)

?? WHO KNEW?

How Many Temperature Scales Are There?

Did you know that in addition to Fahrenheit and Celsius, there is a third temperature scale widely used throughout the world? The Kelvin scale starts at absolute zero, the lowest possible temperature at which there is no heat energy present at all. It is primarily used by scientists to measure very high or very low temperatures when water is not involved.

Converting Between Fahrenheit and Celsius Temperatures

Understanding how to convert between Fahrenheit and Celsius temperatures is an essential skill in understanding metric temperatures. You likely know that below 32 °F means freezing temperatures and perhaps that the same holds true for 0 °C. While it may be difficult to recall that water boils at 212 °F, knowing that it boils at 100 °C is a fairly easy thing to remember.

But what about all the temperatures in between? What is the temperature in degrees Celsius on a scorching summer day? What about a cool autumn afternoon? If a recipe instructs you to preheat the oven to 350 °F, what Celsius temperature do you set the oven at?

[Figure 9.15](#) lists common temperatures on both scales, because we don't use Celsius temperatures daily it's difficult to remember them. Fortunately, we don't have to. Instead, we can convert temperatures from Fahrenheit to Celsius and from Celsius to Fahrenheit using a simple algebraic expression.

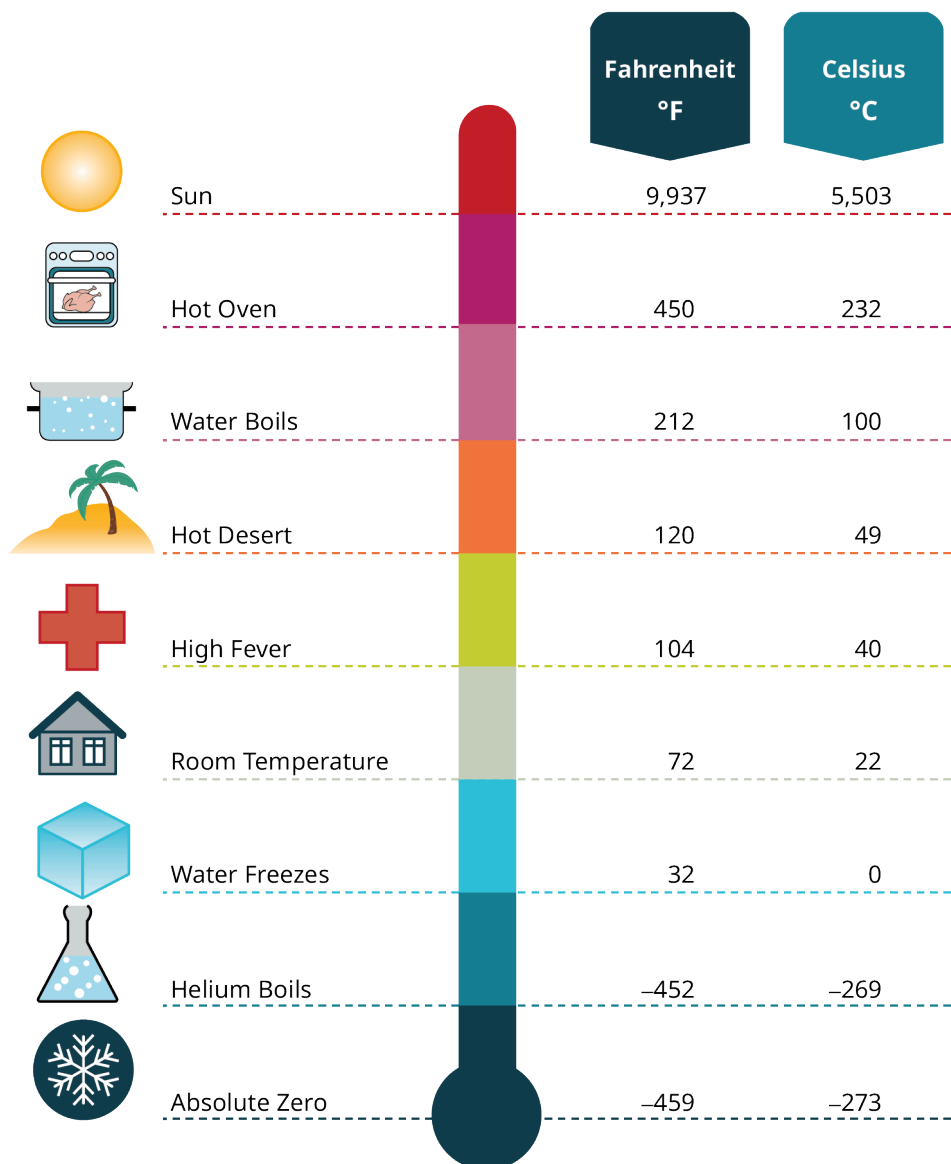


Figure 9.15 Common Temperatures

FORMULA

The formulas used to convert temperatures from Fahrenheit to Celsius or from Celsius to Fahrenheit are outlined in

Table 9.3.

Fahrenheit to Celsius	Celsius to Fahrenheit
$C = \frac{5}{9}(F - 32)$	$F = \frac{9}{5}C + 32$

Table 9.3 Temperature Conversion Formulas

EXAMPLE 9.40**Converting Temperatures from Fahrenheit to Celsius**

A recipe calls for the oven to be set to 392 °F. What is the temperature in Celsius?

✓ Solution

Use the formula in [Table 9.3](#) to convert from Fahrenheit to Celsius.

$$\begin{aligned}
 C &= \frac{5}{9}(F - 32) \\
 C &= \frac{5}{9}(392 - 32) \\
 C &= \frac{5}{9}(360) \\
 C &= 200
 \end{aligned}$$

So, 392 °F is equivalent to 200 °C.

> YOUR TURN 9.40

- What is 482 °F in Celsius?

EXAMPLE 9.41**Converting Temperatures from Celsius to Fahrenheit**

On a sunny afternoon in May, the temperature in London was 20 °C. What was the temperature in degrees Fahrenheit?

✓ Solution

Use the formula in [Table 9.3](#) to convert from Celsius to Fahrenheit.

$$\begin{aligned}
 F &= \frac{9}{5}C + 32 \\
 F &= \frac{9}{5}(20) + 32 \\
 F &= 36 + 32 \\
 F &= 68
 \end{aligned}$$

The temperature was 68 °F.

> YOUR TURN 9.41

- What is 75 °C in Fahrenheit?

EXAMPLE 9.42**Comparing Temperatures in Celsius and Fahrenheit**

A manufacturer requires a vaccine to be stored in a refrigerator at temperatures between 36 °F and 46 °F. The refrigerator in the local pharmacy cools to 3 °C. Can the vaccine be stored safely in the pharmacy's refrigerator?

 Solution

Use the formula in [Table 9.3](#) to convert from Celsius to Fahrenheit.

$$\begin{aligned} F &= \frac{9}{5}C + 32 \\ F &= \frac{9}{5}(3) + 32 \\ F &= 5.4 + 32 \\ F &= 37.4 \end{aligned}$$

Then, compare the temperatures.

$$36\text{ °F} < 37.4\text{ °F} < 46\text{ °F}$$

Yes. 37.4 °F falls within the acceptable range to store the vaccine, so it can be stored safely in the pharmacy's refrigerator.

 YOUR TURN 9.42

1. In July, the average temperature in Madrid is 16.7 °C. The average temperature in Toronto in July is 57.4 °F. Which city has the higher temperature, and by how many degrees Celsius?

Reasonable Values for Temperature

While knowing the exact temperature is important in most cases, sometimes an approximation will do. When trying to assess the reasonableness of values for temperature, there is a quicker way to convert temperatures for an approximation using mental math. These simpler formulas are listed in [Table 9.4](#).

FORMULA

The formulas used to estimate temperatures from Fahrenheit to Celsius or from Celsius to Fahrenheit are outlined in [Table 9.4](#).

Fahrenheit to Celsius	Celsius to Fahrenheit
$C = \frac{F-30}{2}$	$F = 2C + 30$

Table 9.4 Estimate Temperature Conversion

 VIDEO

[Temperature Conversion Trick \(https://openstax.org/r/Temperature_Conversion_Trick\)](https://openstax.org/r/Temperature_Conversion_Trick)

EXAMPLE 9.43**Using Benchmark Temperatures to Determine Reasonable Values for Temperatures**

Which is the more reasonable value for the temperature of a freezer?

- 5 °C or
- -5 °C?

 Solution

We know that water freezes at 0 °C. So, the more reasonable value for the temperature of a freezer is −5 °C, which is below 0 °C. At temperature of 5 °C is above freezing.

 YOUR TURN 9.43

1. To make candy apples, you must boil the sugar mixture for about 20 minutes. Which is the more reasonable value for the temperature of the mixture when boiling:
 - 48 °C or
 - 148 °C?

EXAMPLE 9.44**Using Estimation to Determine Reasonable Values for Temperatures**

The average body temperature is generally accepted as 98.6 °F. What is a reasonable value for the average body temperature in degrees Celsius:

- 98.6 °C,
- 64.3 °C, or
- 34.3 °C?

 Solution

To estimate the average body temperature in degrees Celsius, subtract 30 from the temperature in degrees Fahrenheit, and divide the result by 2.

$$\frac{(98.6 - 30)}{2} = \frac{68.6}{2} = 34.3$$

A reasonable value for average body temperature is 34.3 °C.

 YOUR TURN 9.44

1. Which represents a reasonable value for temperature of a hot summer day:
 - 102 °C,
 - 37 °C, or
 - 20 °C?

EXAMPLE 9.45**Using Conversion to Determine Reasonable Values for Temperatures**

Which is a reasonable temperature for storing chocolate:

- 28 °C,
- 18 °C, or
- 2 °C?

 Solution

Use the formula in [Table 9.3](#) to determine the temperature in degrees Fahrenheit.

$$F = \frac{9}{5}C + 32$$

$$F = \frac{9}{5}(28) + 32 = 82.4$$

$$F = \frac{9}{5}(18) + 32 = 64.4$$

$$F = \frac{9}{5}(2) + 32 = 35.6$$

A temperature of 82.4 °F would be too hot, causing the chocolate to melt. A temperature of 35.6 °F is very close to freezing, which would affect the look and feel of the chocolate. So, a reasonable temperature for storing chocolate is 18 °C, or 64.4 °F.

> YOUR TURN 9.45

- Which represents a reasonable temperature for cooking chili:
 - 240 °C,
 - 60 °C, or
 - 6 °C?

Solving Application Problems Involving Temperature

Whether traveling abroad or working in a clinical laboratory, knowing how to solve problems involving temperature is an important skill to have. Many food labels express sizes in both ounces and grams. Most rulers and tape measures are two-sided with one side marked in inches and feet and the other in centimeters and meters. And while many thermometers have both Fahrenheit and Celsius scales, it really isn't practical to pull out a thermometer when cooking a recipe that uses metric units. Let's review a few instances where knowing how to fluently use the Celsius scale helps solve problems.

EXAMPLE 9.46

Using Subtraction to Solve Temperature Problems

The temperature in the refrigerator is 4 °C. The temperature in the freezer is 21 °C lower. What is the temperature in the freezer?

✓ Solution

Use subtraction to find the difference.

$$4 - 21 = -17$$

So, the temperature in the freezer is -17 °C.

> YOUR TURN 9.46

- At 6 PM, the temperature was 4 °C. By 6 AM the temperature had fallen by 6 °C. What was the temperature at 6 AM?

EXAMPLE 9.47

Using Addition to Solve Temperature Problems

A scientist was using a liquid that was 35 °C. They needed to heat the liquid to raise the temperature by 6 °C. What was the temperature after the scientist heated it?

✓ Solution

Use addition to find the new temperature.

$$35 + 6 = 41$$

The temperature of the liquid was 41 °C after the scientist heated it.

YOUR TURN 9.47

1. A hot dog was cooked in a microwave oven. The hot dog was 4 °C when it was put in the microwave and the temperature increased by 59 °C when it was taken out. What was the temperature of the cooked hot dog?

EXAMPLE 9.48

Solving Complex Temperature Problems

The optimum temperature for a chemical compound to develop its unique properties is 392 °F. When the heating process begins, the temperature of the compound is 20 °C. For safety purposes the compound can only be heated 9 °C every 15 minutes. How long until the compound reaches its optimum temperature?

Solution

Step 1: Determine the optimum temperature in degrees Celsius using the formula in [Table 9.3](#).

$$C = \frac{5}{9}(F - 32)$$

$$C = \frac{5}{9}(392 - 32)$$

$$C = \frac{5}{9}(360)$$

$$C = 200$$

Step 2: Subtract the starting temperature.

$$200\text{ }^{\circ}\text{C} - 20\text{ }^{\circ}\text{C} = 180\text{ }^{\circ}\text{C}$$

Step 3: Determine the number of 15-minute cycles needed to heat the compound to its optimum temperature.

$$180 \div 9 = 20$$

Step 4: Multiply the number of cycles needed by 15 minutes and convert the product to hours and minutes.

$$15\text{ minutes} \times 9 = 135\text{ minutes}$$

$$135\text{ minutes} = 2\text{ hours } 15\text{ minutes}$$

So, it will take 2 hours and 15 minutes for the compound to reach its optimum temperature.

YOUR TURN 9.48

1. After reaching a temperature of 302 °F a chemical compound cools at the rate of 5 °C every 6 minutes. How long will it take until the compound has reached a temperature of zero degree Celsius.

VIDEO

[Learn the Metric System in 5 Minutes \(https://openstax.org/r/Metric_System_in_5_Minutes\)](https://openstax.org/r/Metric_System_in_5_Minutes)

Check Your Understanding

For the following exercises, determine the most reasonable value for each temperature.

34. Popsicle:
28.5 °C or

- 28.5 °F
35. Room temperature:
20 °F or
20 °C
36. Hot coffee:
71.1 °C or
71.1 °F

For the following exercises, use mental math to approximate the temperature.

37. 450 °F = _____ °C
38. 35 °C = _____ °F
39. 100 °F = _____ °C

For the following exercises, convert the temperatures to the nearest degree.

40. 225 °F = _____ °C
41. 27 °C = _____ °F
42. 750 °F = _____ °C



SECTION 9.5 EXERCISES

For the following exercises, determine the most reasonable value for each temperature.

- Human body:
37 °C or
37 °F
- An ice cube:
32 °C or
0 °C
- Boiling water:
212 °C
or 212 °F
- Summer day:
8 °C,
23 °C, or
75 °C
- Winter day:
33 °C,
30 °C, or
3 °C
- Lava:
700 °C,
70 °C, or
7 °C

For the following exercises, use mental math to approximate the temperature.

- 500 °F = _____ °C
- 25 °C = _____ °F
- 350 °F = _____ °C
- 150 °C = _____ °F
- 4 °F = _____ °C
- 73 °C = _____ °F
- 72 °F = _____ °C
- 10 °C = _____ °F
- 1,020 °F = _____ °C

For the following exercises, convert the temperatures to the nearest degree.

- 450 °F = _____ °C
- 35 °C = _____ °F
- 525 °F = _____ °C
- 140 °C = _____ °F
- 40 °F = _____ °C

21. $67^{\circ}\text{C} = \underline{\hspace{1cm}}^{\circ}\text{F}$
22. $85^{\circ}\text{F} = \underline{\hspace{1cm}}^{\circ}\text{C}$
23. $15^{\circ}\text{C} = \underline{\hspace{1cm}}^{\circ}\text{F}$
24. $1,200^{\circ}\text{F} = \underline{\hspace{1cm}}^{\circ}\text{C}$
25. $112^{\circ}\text{F} = \underline{\hspace{1cm}}^{\circ}\text{C}$
26. $105^{\circ}\text{C} = \underline{\hspace{1cm}}^{\circ}\text{F}$
27. $125^{\circ}\text{F} = \underline{\hspace{1cm}}^{\circ}\text{C}$
28. A cup of tea is 30°C . After adding ice, the temperature of the tea decreased 3°C . What is the temperature of the tea now?
29. A liquid with a temperature of 15°C is placed on a stovetop. The liquid is heated at the rate of 1.5°C per minute. What is the temperature of the liquid after 10 minutes?
30. The temperature inside a store is 22°C . Outside the temperature is 37°C . How much cooler is it in the store than outside?
31. A boiling pan of water is 212°F . As the water cools the temperature drops 6°C every 2 minutes. How many minutes until the temperature of the water reaches 40°C ?
32. On May 1 the temperature was 79°F . On June 1 the temperature was 25°C . Which day was warmer?
33. For each log placed on a fire, the temperature increases 45°C . How many logs are needed for the campfire to increase 315°C ?
34. The temperature of a Bunsen burner flame increases 572°F each minute. About how many minutes does it take for the flame to increase $1,200^{\circ}\text{C}$?
35. The instructions to cook a pizza say to set the oven at 425°F . To the nearest degree, what is the temperature in degrees Celsius?
36. In the evening the temperature was 3°C . By morning, the temperature had fallen 4°C . What is the temperature now?
37. An 8,000 BTU air conditioner can cool a room 1°C every 5 minutes. If the temperature in the room is 24°C , how long will it take the air conditioner to cool the room to 20°C ?
38. What is 95°F in degrees Celsius?
39. What is 50°C in degrees Fahrenheit?
40. Which is a reasonable value for the temperature of a room:
 - 60°C ,
 - 40°C , or
 - 20°C ?